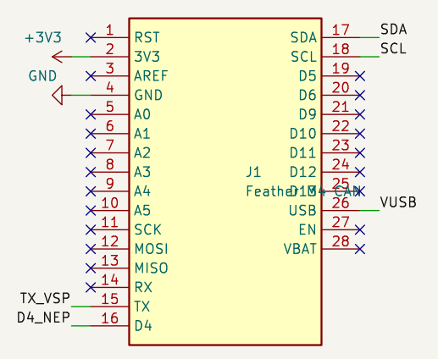
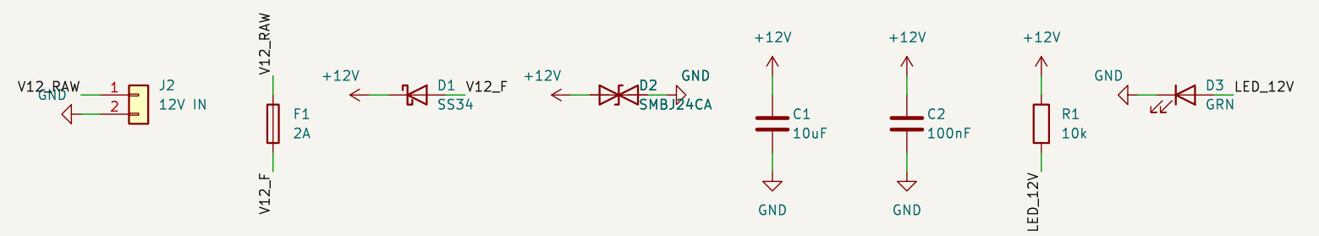
Bench stimulus FeatherWing: 12 V power, SAS power + CAN pass-through, EPS NEP/VSP/torque stimulus

Every EPS line goes through a relay whose NC contact is the car's own wire: no power, no ARM loop or
no enable bit means the EPS sees the car exactly as without this board. ARM loop on J5 (pins 7–8): wire
an NC e–stop or key switch. Faking VT3/VT6 makes the motor steer: wheels off the ground, hands clear.
All enables come from U4 (TCA9534), which powers up with every pin an input; 100k pull–downs hold
every driver off until firmware sets the bit.

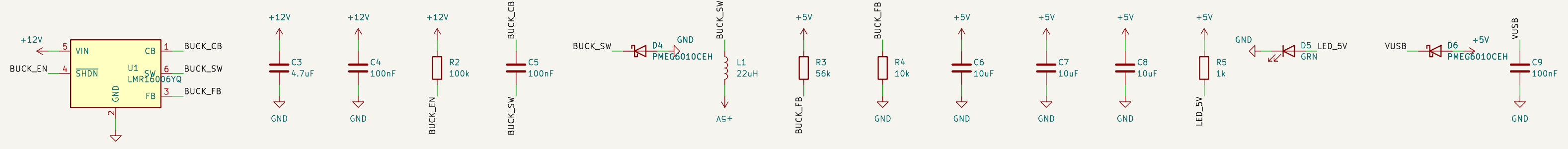
Feather M4 CAN (stacking headers)



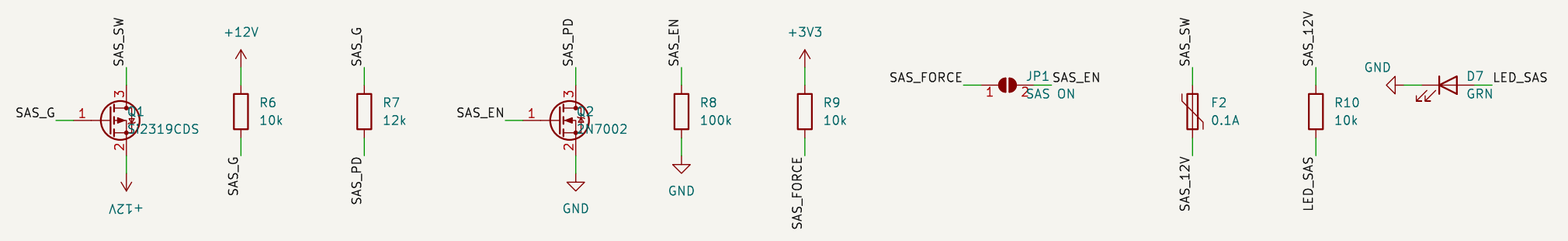
12 V input: fuse, reverse–polarity diode, load–dump TVS



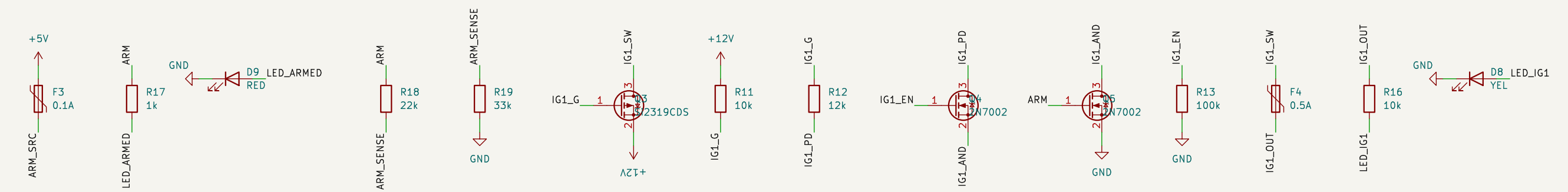
5 V buck (LMR16006, 60 V) and 5 V into the Feather USB pin



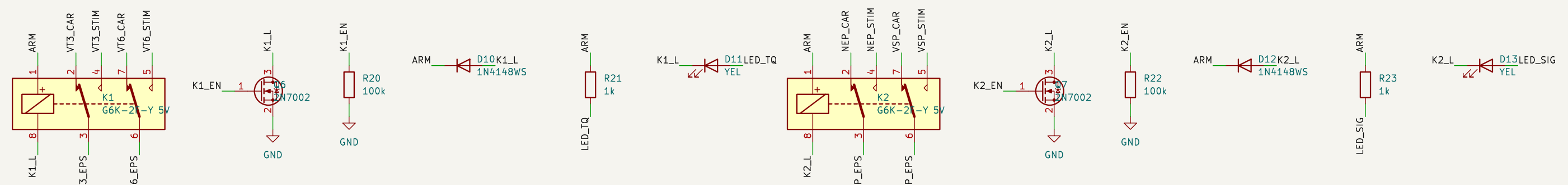
SAS power (12 V, default off; JP1 = always on)



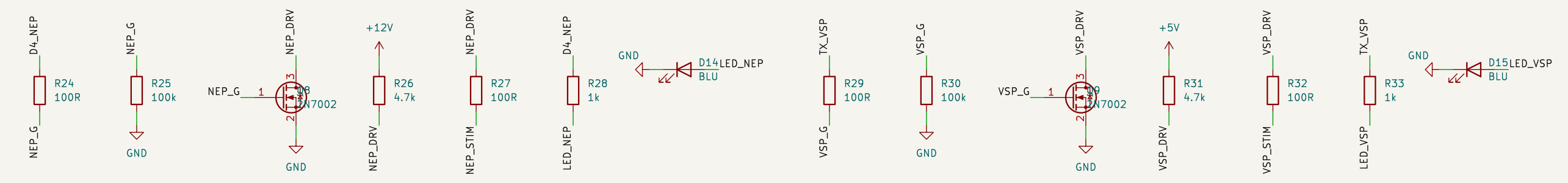
ARM loop (hard kill) and IG1 output (needs IG1_EN and ARM)



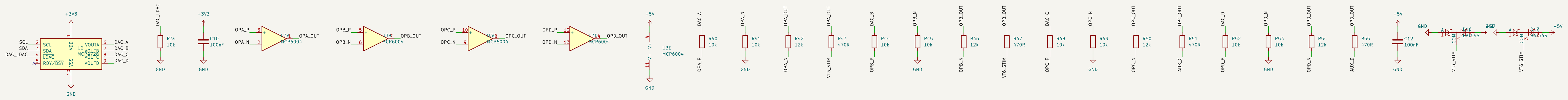
Relays: de–energised = car wiring connected



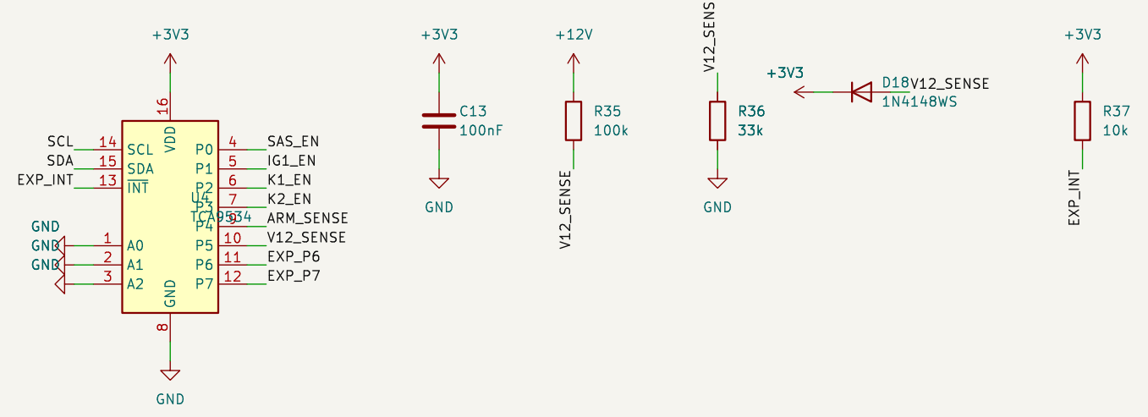
NEP (12 V pull–up) and VSP (5 V pull–up) open–drain outputs



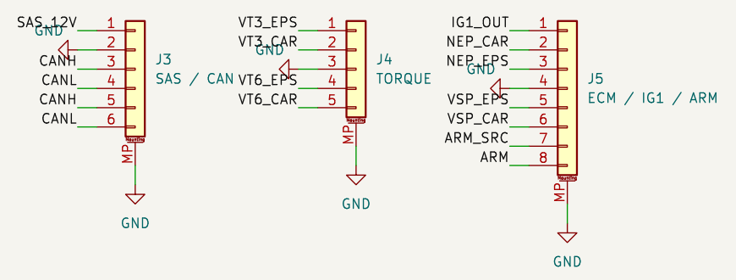
Torque DAC: MCP4728 (2.048 V ref, x1) –> MCP6004 x2.2 on 5 V –> 0–4.5 V



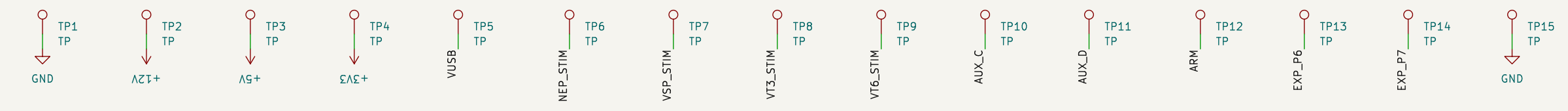
I2C expander TCA9534 (0x20): powers up as inputs



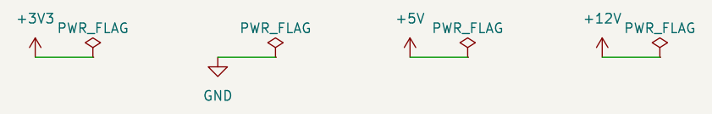
Connectors



Test pads



Power: +3V3/GND from the Feather; +12V via D1; +5V from the buck (L1)



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Honda Insight EPS bench / in–car stimulus + 12 V power

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File: bench_stim_wing.kicad_sch

Title: Bench stimulus FeatherWing

Size: A0 Date: 2026–11–02 **Rev: 1.0**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4 Id: 1/1